08. Model Evaluation

Interpret the available sample outputs and the challenge metrics without overclaiming.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Challenge metric formula

Overall score = 0.4 × F1(theme classification) + 0.2 × accuracy(brand identification) + 0.2 × F1(sentiment) + 0.2 × average headline similarity. The task statement also names precision/recall/F1 for theme and sentiment, and ROUGE/BLEU plus embedding similarity for headlines.

What the notebook reports

Stage	Observed output / timing
Evaluation sample	222 articles and 210 tweets initially displayed
Predicted mobile-tech subset	71 articles; 62 tweets
Language detection	Articles: 45 en, 24 hi, 2 hing; tweets: 32 en, 18 hi, 12 hing
Headline generation	T5 generated headline strings for the mobile-tech article subset
Sentiment output	557 brand-level rows exported
Total runtime	361.98 seconds in the saved notebook run

Runtime breakdown

Stage	Seconds
Initial cleaning + regex brands	1.07
Binary theme classification	0.20
Language detection	4.47
Second preprocessing / brand extraction	0.25
Translation	290.37
T5 headline generation	60.87
Brand context preprocessing	1.12
Sentiment inference	3.63
Total	361.98

Interpretation

Translation dominates the recorded runtime (about 80% of the total), while headline generation is the second-largest component. This supports the report's stated bottleneck. These timings come from one historical Colab run and should not be treated as current service-level latency.

Metric explanations

- Precision: fraction of predicted positives that are correct; useful for controlling false alarms.
- Recall: fraction of actual positives recovered; useful for avoiding missed mobile-tech content or brand sentiment labels.
- F1: harmonic mean of precision and recall.
- Brand identification accuracy: exact brand entity recognition correctness under the challenge's scoring definition.
- ROUGE: n-gram overlap with reference headline; BLEU: precision-oriented n-gram overlap; embedding similarity: semantic closeness.

Evaluation limitations

- The supplied notebook outputs do not include numeric precision/recall/F1 for theme or sentiment, brand-identification accuracy, or ROUGE/BLEU headline scores.
- A generated headline example is not a quantitative evaluation.
- The evaluation sample row counts vary between displayed dataframe shape and later class counts; reconcile before reporting exact denominators.
- The report's challenge formula is a target metric, not evidence of the project's achieved final score.